

Linear Inequalities: Solving and Graphing

Theory + 30 practice questions (10 easy · 10 medium · 5 medium-hard · 5 hard) with full step-by-step solutions. Class 3 of the Young Fermats Algebra Ignite course at SOMATH — the School of Math on the Upper West Side of NYC.

How to use this packet

Work each question on scratch paper first, then check the solution at the back. Every solution follows the same 4 rules: Isolate · Flip when needed · Endpoint choice · Test point. The goal is not to memorize the answer — it is to build the reflex of flipping the sign when you multiply or divide by a negative.

The 4 rules of inequalities

1. Isolate. Solve like an equation — add, subtract, multiply, divide the same thing to both sides.
2. Flip on negative. If you multiply or divide by a negative number, flip the inequality symbol. This is the #1 source of Regents inequality errors.
3. Endpoint. Strict ($<$, $>$) → open circle or dashed line. Or-equal ($\leq$, $\geq$) → closed dot or solid line.
4. Test point. Pick any number in your shaded region and plug into the original. If it makes a true statement, you shaded correctly.

Practice Questions

Work these first. Full solutions start on the solutions page.

Section 1 · Easy (Questions 1–10)

Question 1 · Easy

Solve for x : $x + 4 > 9$.

Question 2 · Easy

Solve for x : $x - 7 \leq -2$.

Question 3 · Easy

Solve for x : $3x < 15$.

Question 4 · Easy

Solve for x : $-2x < 10$.

Question 5 · Easy

Solve for x : $2x + 5 \geq 13$.

Question 6 · Easy

Solve for x : $4 - x > 1$.

Question 7 · Easy

Solve for x : $x/3 \leq 4$.

Question 8 · Easy

Solve for x : $-x/4 \geq 2$.

Question 9 · Easy

Solve for x : $5x - 3 < 2x + 6$.

Question 10 · Easy

Write the inequality shown on this number line: closed dot at -2 , arrow going right.

Section 2 · Medium (Questions 11–20)

Question 11 · Medium

Solve for x : $3(x - 2) < 12$.

Question 12 · Medium

Solve for x : $-4(2x + 1) \geq 20$.

Question 13 · Medium

Solve for x : $(x + 3)/2 > 5$.

Question 14 · Medium

Solve for x : $6 - 2x \leq 4x - 12$.

Question 15 · Medium

Solve the compound inequality: $-1 < 2x + 3 \leq 9$.

Question 16 · Medium

Solve the compound inequality: $x - 4 < -5$ OR $x - 4 \geq 1$.

Question 17 · Medium

Solve for x : $|x| \leq 6$.

Question 18 · Medium

Solve for x : $|x - 2| > 5$.

Question 19 · Medium

Write the interval-notation solution to $x < 4$ AND $x \geq -1$.

Question 20 · Medium

Solve for x : $2x/3 - 1 < 5$.

Section 3 · Medium–Hard (Questions 21–25)

Question 21 · Medium-Hard

Solve for x : $(x + 1)/3 - (x - 2)/4 \geq 1$.

Question 22 · Medium-Hard

Solve for x : $|3x + 5| < 11$.

Question 23 · Medium-Hard

A ride-share app charges a \$3 base fee plus \$2.50 per mile. Nora has at most \$18 to spend. Write and solve an inequality for the greatest whole number of miles she can travel.

Question 24 · Medium-Hard

Graph the solution of $y < 2x - 3$ in the coordinate plane. Describe the boundary and which region is shaded.

Question 25 · Medium-Hard

Solve for x : $5 - 3(2x - 1) > 2(x + 4) - 5$.

Section 4 · Hard (Questions 26–30)**Question 26 · Hard**

Solve for x : $2|x - 4| + 3 < 11$.

Question 27 · Hard

Determine all values of x that satisfy $|2x + 3| \geq -4$.

Question 28 · Hard

Solve the system in the coordinate plane and describe the solution region: $y \geq x + 1$ AND $y < -2x + 4$.

Question 29 · Hard

Sasha earns \$12 per hour babysitting and \$8 per hour dog-walking. She wants to earn at least \$120 this week and can work no more than 15 hours total. Write a system of inequalities for the number of babysitting hours b and dog-walking hours d , and describe one combination that works.

Question 30 · Hard

Solve for x : $|4x - 7| < 2x + 1$.

Solutions

Full step-by-step reasoning for every question.

Question 1 · Easy

Solve for x : $x + 4 > 9$.

Answer: $x > 5$

Subtract 4 from both sides. No flip needed (we only added/subtracted). Interval: $(5, \infty)$. Open circle at 5, arrow right.

Question 2 · Easy

Solve for x : $x - 7 \leq -2$.

Answer: $x \leq 5$

Add 7 to both sides. Interval: $(-\infty, 5]$. Closed dot at 5, arrow left.

Question 3 · Easy

Solve for x : $3x < 15$.

Answer: $x < 5$

Divide both sides by 3. Positive divisor — no flip. Interval: $(-\infty, 5)$.

Question 4 · Easy

Solve for x : $-2x < 10$.

Answer: $x > -5$

Divide both sides by -2 . Negative divisor — flip the $<$ to $>$. Interval: $(-5, \infty)$. This is the classic sign-flip problem.

Question 5 · Easy

Solve for x : $2x + 5 \geq 13$.

Answer: $x \geq 4$

Subtract 5: $2x \geq 8$. Divide by 2: $x \geq 4$. Interval: $[4, \infty)$.

Question 6 · Easy

Solve for x : $4 - x > 1$.

Answer: $x < 3$

Subtract 4: $-x > -3$. **Multiply both sides by -1 and flip:** $x < 3$. **Interval:** $(-\infty, 3)$.

Question 7 · Easy

Solve for x : $x/3 \leq 4$.

Answer: $x \leq 12$

Multiply both sides by 3. Positive multiplier — no flip. **Interval:** $(-\infty, 12]$.

Question 8 · Easy

Solve for x : $-x/4 \geq 2$.

Answer: $x \leq -8$

Multiply both sides by -4 and flip the $\geq$ to $\leq$. **Interval:** $(-\infty, -8]$.

Question 9 · Easy

Solve for x : $5x - 3 < 2x + 6$.

Answer: $x < 3$

Subtract $2x$: $3x - 3 < 6$. **Add 3:** $3x < 9$. **Divide by 3:** $x < 3$. **Interval:** $(-\infty, 3)$.

Question 10 · Easy

Write the inequality shown on this number line: closed dot at -2 , arrow going right.

Answer: $x \geq -2$

Closed dot $\rightarrow$ "or equal to." **Arrow right $\rightarrow$ greater than.** **Interval:** $[-2, \infty)$. **Set-builder:** $\{x \mid x \geq -2\}$.

Question 11 · Medium

Solve for x : $3(x - 2) < 12$.

Answer: $x < 6$

Distribute: $3x - 6 < 12$. **Add 6:** $3x < 18$. **Divide by 3:** $x < 6$. **Interval:** $(-\infty, 6)$.

Question 12 · Medium

Solve for x: $-4(2x + 1) \geq 20$.

Answer: $x \leq -3$

Distribute: $-8x - 4 \geq 20$. Add 4: $-8x \geq 24$. Divide by -8 and flip: $x \leq -3$. Interval: $(-\infty, -3]$.

Question 13 · Medium

Solve for x: $(x + 3)/2 > 5$.

Answer: $x > 7$

Multiply both sides by 2 (positive, no flip): $x + 3 > 10$. Subtract 3: $x > 7$. Interval: $(7, \infty)$.

Question 14 · Medium

Solve for x: $6 - 2x \leq 4x - 12$.

Answer: $x \geq 3$

Add 2x to both sides: $6 \leq 6x - 12$. Add 12: $18 \leq 6x$. Divide by 6: $3 \leq x$, i.e. $x \geq 3$. Interval: $[3, \infty)$.

Question 15 · Medium

Solve the compound inequality: $-1 < 2x + 3 \leq 9$.

Answer: $-2 < x \leq 3$

Subtract 3 from every part: $-4 < 2x \leq 6$. Divide every part by 2: $-2 < x \leq 3$. Interval: $(-2, 3]$.

Open circle at -2 , closed dot at 3 , shade between.

Question 16 · Medium

Solve the compound inequality: $x - 4 < -5$ OR $x - 4 \geq 1$.

Answer: $x < -1$ OR $x \geq 5$

Solve each side separately. Left: $x < -1$. Right: $x \geq 5$. Interval: $(-\infty, -1) \cup [5, \infty)$. Two arrows going outward.

Question 17 · Medium

Solve for x: $|x| \leq 6$.

Answer: $-6 \leq x \leq 6$

"Less than" absolute value $\rightarrow$ sandwich. Rewrite as $-6 \leq x \leq 6$. Interval: $[-6, 6]$.

Question 18 · Medium

Solve for x : $|x - 2| > 5$.

Answer: $x < -3$ OR $x > 7$

"Greater than" absolute value → two arrows. Split: $x - 2 < -5$ OR $x - 2 > 5$. **First:** $x < -3$.

Second: $x > 7$. **Interval:** $(-\infty, -3) \cup (7, \infty)$.

Question 19 · Medium

Write the interval-notation solution to $x < 4$ AND $x \geq -1$.

Answer: $[-1, 4)$

AND → intersection. The overlap is $-1 \leq x < 4$. **Closed bracket at** -1 (from $\geq$), **open parenthesis at** 4 (from strict $<$).

Question 20 · Medium

Solve for x : $2x/3 - 1 < 5$.

Answer: $x < 9$

Add 1: $2x/3 < 6$. **Multiply by** $3/2$ (positive, no flip): $x < 9$. **Interval:** $(-\infty, 9)$.

Question 21 · Medium-Hard

Solve for x : $(x + 1)/3 - (x - 2)/4 \geq 1$.

Answer: $x \geq 2$

Multiply both sides by the LCD 12 (positive, no flip): $4(x + 1) - 3(x - 2) \geq 12$. **Distribute:** $4x + 4 - 3x + 6 \geq 12$. **Combine:** $x + 10 \geq 12$. **Subtract 10:** $x \geq 2$. **Interval:** $[2, \infty)$.

Question 22 · Medium-Hard

Solve for x : $|3x + 5| < 11$.

Answer: $-16/3 < x < 2$

"Less than" absolute value → sandwich. Rewrite as $-11 < 3x + 5 < 11$. **Subtract 5 from every part:** $-16 < 3x < 6$. **Divide every part by 3 (positive):** $-16/3 < x < 2$. **Interval:** $(-16/3, 2)$.

Question 23 · Medium-Hard

A ride-share app charges a \$3 base fee plus \$2.50 per mile. Nora has at most \$18 to spend. Write and solve an inequality for the greatest whole number of miles she can travel.

Answer: 6 miles

Let m = miles. Then $3 + 2.5m \leq 18$. Subtract 3: $2.5m \leq 15$. Divide by 2.5: $m \leq 6$. The greatest whole-mile trip she can afford is 6 miles.

Question 24 · Medium-Hard

Graph the solution of $y < 2x - 3$ in the coordinate plane. Describe the boundary and which region is shaded.

Answer: Dashed boundary line $y = 2x - 3$; shade the region below the line.

The boundary is $y = 2x - 3$ (slope 2, y-intercept -3). Strict $< \rightarrow$ dashed. Since the inequality is $y <$, shade below. Check with test point $(0, 0)$: $0 < 2(0) - 3 = -3$? No, $0 < -3$ is false. So $(0, 0)$ is not in the solution — it's above the line, and we shaded below. Correct.

Question 25 · Medium-Hard

Solve for x : $5 - 3(2x - 1) > 2(x + 4) - 5$.

Answer: $x < 5/8$

Distribute both sides: $5 - 6x + 3 > 2x + 8 - 5$. Simplify: $8 - 6x > 2x + 3$. Subtract $2x$: $8 - 8x > 3$. Subtract 8: $-8x > -5$. Divide by -8 and flip: $x < 5/8$. Interval: $(-\infty, 5/8)$.

Question 26 · Hard

Solve for x : $2|x - 4| + 3 < 11$.

Answer: $0 < x < 8$

Isolate the absolute value first. Subtract 3: $2|x - 4| < 8$. Divide by 2: $|x - 4| < 4$. Now the sandwich: $-4 < x - 4 < 4$. Add 4 to every part: $0 < x < 8$. Interval: $(0, 8)$.

Question 27 · Hard

Determine all values of x that satisfy $|2x + 3| \geq -4$.

Answer: All real numbers, $(-\infty, \infty)$

Absolute value is always ≥ 0 , and $0 \geq -4$ is automatically true. So every real number x satisfies this inequality. This is the “all real numbers” case — don't waste time splitting; recognize it and write the answer.

Related trap: If the problem had been $|2x + 3| \leq -4$, the answer would be no solution (an absolute value is never less than a negative). Always look at the right side of an absolute-value inequality before you split.

Question 28 · Hard

Solve the system in the coordinate plane and describe the solution region: $y \geq x + 1$ AND $y < -2x + 4$.

Answer: The wedge-shaped region above the solid line $y = x + 1$ and below the dashed line $y = -2x + 4$. The two boundaries meet at $(1, 2)$; the solution is bounded on the left by their intersection.

Line 1: $y = x + 1$, solid (because $\geq$), shade above. Line 2: $y = -2x + 4$, dashed (because strict $<$), shade below. The overlap is a wedge to the left of the intersection point. Solve the system to find the vertex: set $x + 1 = -2x + 4 \rightarrow 3x = 3 \rightarrow x = 1$, $y = 2$. Test point $(0, 3)$: first, $3 \geq 0 + 1 = 1 \checkmark$. Second, $3 < -2(0) + 4 = 4 \checkmark$. So $(0, 3)$ is in the region — that confirms the wedge to the left.

Question 29 · Hard

Sasha earns \$12 per hour babysitting and \$8 per hour dog-walking. She wants to earn at least \$120 this week and can work no more than 15 hours total. Write a system of inequalities for the number of babysitting hours b and dog-walking hours d , and describe one combination that works.

Answer: $12b + 8d \geq 120$, $b + d \leq 15$, $b \geq 0$, $d \geq 0$. One valid combination: $b = 10$, $d = 0$ (earns \$120, works 10 hours).

Total pay: $12b + 8d \geq 120$ (“at least \$120”). Total hours: $b + d \leq 15$ (“no more than 15 hours”). Non-negativity: hours can't be negative. The solution region in the (b, d) -plane is bounded by these four inequalities. Any lattice point in the region is a valid combination. Another example: $b = 6$, $d = 6$ (pay \$72 + \$48 = \$120 $\checkmark$, hours 12 $\checkmark$).

Question 30 · Hard

Solve for x : $|4x - 7| < 2x + 1$.

Answer: $1 < x < 4$

This is the trickiest form: the right side is a variable expression, not a constant. Two things must be true. First, $2x + 1 > 0$ (otherwise the absolute value, which is ≥ 0 , cannot be less than a non-positive number). That gives $x > -1/2$. Second, apply the "less than" sandwich: $-(2x + 1) < 4x - 7 < 2x + 1$.

Right piece: $4x - 7 < 2x + 1 \rightarrow 2x < 8 \rightarrow x < 4$.

Left piece: $-(2x + 1) < 4x - 7 \rightarrow -2x - 1 < 4x - 7 \rightarrow 6 < 6x \rightarrow x > 1$.

Combine with $x > -1/2$ from the domain check: the intersection is $1 < x < 4$. Interval: $(1, 4)$.

Test point $(0, 0)$ is not in the interval: check original — $|4(0) - 7| = 7$, and $2(0) + 1 = 1$; $7 < 1$ is false, as expected. Test $x = 2$ (should be in): $|4(2) - 7| = 1$, $2(2) + 1 = 5$; $1 < 5$ is true. Correct.

Answer Key Summary

All 30 answers at a glance.

#	Level	Answer
1	Easy	$x > 5$
2	Easy	$x \leq 5$
3	Easy	$x < 5$
4	Easy	$x > -5$ (flip!)
5	Easy	$x \geq 4$
6	Easy	$x < 3$ (flip!)
7	Easy	$x \leq 12$
8	Easy	$x \leq -8$ (flip!)
9	Easy	$x < 3$
10	Easy	$x \geq -2$
11	Medium	$x < 6$
12	Medium	$x \leq -3$ (flip!)
13	Medium	$x > 7$
14	Medium	$x \geq 3$
15	Medium	$-2 < x \leq 3$
16	Medium	$x < -1$ OR $x \geq 5$
17	Medium	$-6 \leq x \leq 6$
18	Medium	$x < -3$ OR $x > 7$
19	Medium	$[-1, 4)$
20	Medium	$x < 9$
21	Medium-Hard	$x \geq 2$
22	Medium-Hard	$-16/3 < x < 2$
23	Medium-Hard	6 miles
24	Medium-Hard	Dashed $y = 2x - 3$; shade below
25	Medium-Hard	$x < 5/8$ (flip!)
26	Hard	$0 < x < 8$
27	Hard	All real numbers

#	Level	Answer
28	Hard	Wedge above solid $y = x + 1$ and below dashed $y = -2x + 4$
29	Hard	$12b + 8d \geq 120$, $b + d \leq 15$, $b, d \geq 0$
30	Hard	$1 < x < 4$

About SOMATH

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